

AUD213

Basic Mixing Concepts for the Screen

[AUD213](#)

[Basic Mixing Concepts for the Screen](#)

[Basic Mixing Concepts](#)

[Mixing Workflow](#)

[Evaluating the Mix](#)

[MIX PROCESS](#)

[Balancing and Equalisation](#)

[Premixing](#)

[Final Mix](#)

[Printmastering](#)

[Delivery items](#)

[Dynamics, EQ, Effects, Placement & Panning](#)

[DYNAMICS & EQ](#)

[Compression](#)

[Limiting](#)

[Gate](#)

[EQ \(Frequency\)](#)

[De-esser](#)

[EQ Tips](#)

[EFFECTS](#)

[Reverb](#)

[Delay/Echo](#)

[Modulation Effects](#)

[PLACEMENT and PANNING](#)

[Dimension](#)

Basic Mixing Concepts

Mixing sound for film is a complex and wonderful art.

The Mixer/s, Editors, Composer, and Director work together to balance and refine the soundscape for the film, not just in terms of an aural nature, but also in terms of the most suitable storytelling techniques a film requires.

The Sound will be broken up into Separate categories for pre-mixing

- ❖ Dialogue
- ❖ ADR
- ❖ Fx
- ❖ Atmospheres
- ❖ Foley
- ❖ Music

At any given time the sound mix will be designed using a combination of the following elements and storytelling techniques

- ❖ Balance
- ❖ Frequency Range
- ❖ Panorama
- ❖ Dimension
- ❖ Dynamics Psycho-acoustics Principles
- ❖ Synchronous versus Asynchronous Sounds

Mixing Workflow

The Mixer will work with the separate elements of the soundtrack in an orderly manner. For example, they will mix Atmospheres, Dialogue, ADR, Foley, and Fx, and create separate Premixes.

(The order of Premixing is dependent on the individual mixers' preference – Some mixers prefer to start with the Atmospheres to set the general bed, tone, and base of the mix bed, then mix the dialogues, fx, etc., and some prefer to start with Dialogues. Every production is unique and discussion with the Mixer about their preference is essential. Scheduling issues and finance are also a consideration as per usual.)

Premixes are created by 'folding down' the separated and sometimes large number of tracks delivered by the sound editor into a more manageable form.

The Mixer will balance, equalise, pan, and “clean up” the sounds and create a number of premixes for each of the elements (Dial, Fx, Atmos, Foley) to keep sounds separated enough so that they can have complete control in case the Director wants to change the level, effect or delete certain sounds.

The Dial, Fx, Atmos, and Foley premixes should be mixed so they can be played at unity or 0 across the mixing board sounding as good as possible before the Final mix begins.

The **Music mixes** are usually delivered during the premixing stage as either STEMs or Mixed files, and the mixer will play them as a guide against the Dial, Fx, Atmos, & Foley to get a sense of how the soundtrack as a whole will work together.

At this stage, it is common to have a **Premix Screening**.

This is a double head continuous screening of the film (with the workprint image / or digital payout) in a cinema-like environment or a continuous playback screening in a television mix room.

The premix screening gives the mixer, editor, and sound supervisor/editors the opportunity to get an overview of the sonic balance, scene transitions and to **make note** of any synchronisation issues, music, fx or dial balance problems, identify any sound that is missing and to see and hear what is working nicely together or negatively for the narrative, both acoustically and psychoacoustically.

It is also the opportunity to make sure that the aural soundscape is in accordance with the director’s vision, and the creative aesthetic decisions that were discussed.

Once the **premix notes** have been established, [fix ups completed - if any] the **final mixing** can begin.

All premixes are played simultaneously, and it is at this stage the fun really begins. It’s a process of balancing, fine-tuning, and blending all the elements together. Sometimes it is a process of simplifying the soundtrack, deleting some sounds in order to make way for others to be more predominant in the sound frequency spectrum.

Mixing is about light & shade, balance & perspective, and working intuitively, emotionally, and technically with sound to enhance the narrative and work dynamically with the pictures on the screen.

The Director of the film will usually be present throughout the Final Mix stage, although sometimes the sound supervisor will be the spokesperson for the director to make sure the director's vision is realised.

As with all other arenas in filmmaking, the mix is a collaborative process with each artist ie. Mixer, Picture Editor, Sound Editor, Composer, Director, and Producer working together to fulfill a unified vision.

Diplomacy and cooperation are vital to the process.

Upon completion of the Final Mix a continuous screening of the film or programme, called a **Playback or Final Mix Screening** will take place.

Fixups or changes will be duly **noted** and then the final mix will be changed according to the fix-up notes.

The **Final Mix** is also separated into STEMs called a **DME** (Dialogue, Music, and Effects).

In Film, the final stage is the **Print Master**. The printmaster is where the digital mastering process of the DME is combined to create a final composite track.

When this is completed, an optical or digital soundtrack can be created for a feature film **release print**.

In Television, the final stage is the **Mix Delivery** as a digital file, in Stereo or 5.1.

Typically you would provide the following:

- ❖ Full Mix 5.1
- ❖ Full Mix Stereo
- ❖ SEPARATE STEMs: DME (DIALOGUE, MUSIC, AND EFFECTS)
- ❖ SEPARATE STEMs: M&E (MUSIC AND EFFECTS)

Delivery items (As per the Producers Contractual Delivery Schedule) are made from the separate Final mix stems ie:(D, M & E), for various international (M&E), domestic (English language), Airline, and T.V. versions. Normally made directly after the **Printmaster / Final Mix** stage and completed by the mixer.

Evaluating the Mix

Factors to be considered when evaluating your mix:

- ❖ **Delivery format:** Before you begin any production, ascertain what format of delivery system the program will be reproduced; ie Film, TV, game, web distribution etc, as all of the above, have different sonic and technical specifications for delivery.
- ❖ **Source Quality:** The original recordings of sound must be the best possible quality.
- ❖ **Intelligibility:** Narration, Dialogue, and Song Lyrics must be intelligible. Be aware that when working with program material over a long period of time, words become so familiar that it might not be apparent that they are muffled, masked, or difficult to distinguish. If you are unsure, ask someone who is not familiar with the program to listen with “Fresh Ears”. Also, look back on your 1st reaction/ gut reaction/ intuited notes- this will help you remember phrases, etc that were not clear on 1st viewing. Remember...the audience only gets to listen once.
- ❖ **Tone, Spatial balance, and Perspective:** Sound elements are spread across the frequency spectrum, to avoid blurring, and muddiness. It should also be clear which direction sounds are coming from. Spatial relationships- front to back and side to side, should be in proper perspective. ie; a background crowd should not be louder than CU dialogue, a distant train coming towards you should become louder and more defined as it approaches.
- ❖ **Definition:** Each element should be clearly defined. Identifiable, separate, and distinct, yet if grouped they should be blended so that no one element stands out or crowds or masks another sound. Each element should have its position in, and yet be part of the overall spectral range and spatial arrangement. In a sense, you are recreating Aural normality.

- ❖ **Dynamic Range:** The range of sounds from softest to loudest should be as wide as the medium allows. The Softest sounds should be easily audible and the loudest sounds are undistorted.
- ❖ **Clarity:** Sounds should be noise and distortion-free. Hum, hiss, leakage, phasing, smearing, blurring from too much reverb, harmonic dysfunction, and loudness distortion – all muddle sound, adversely affecting the clarity.
- ❖ **Airiness:** The soundtrack should be airy and open. It should not sound closed down, muffled, constricted, lifeless, or oppressive unless that is the desired creative result.
- ❖ **Acoustical Appropriateness:** The space in which a character is seen and the dimension of that space should match. ie: Hall, small room, tunnel.

Establish the MAIN and SUPPORTING sounds to create focus and POV (Point of View). Position sounds to create relationships of space and distance and maintain balance so the aural space is properly weighted. Sound elements should have different sonic features to maintain definition and Intelligibility; ie Tempo, pitch, loudness, timbre, and style.

MIX PROCESS

This is the first time when the Editor's work is displayed (naked) when all tracks are played separately for the Mixer and Director.

It can be an overwhelming experience if it is your first editorial attempt, however, if you have presented clean and concise tracks and have been thoughtful in your approach to tracklaying it is a wonderful learning experience.

Balancing and Equalisation

- ☐ All sounds balanced and blended [using clip gain], at levels that work with one another and with the image

- ❑ Cleaned up and equalised to reduce noise (inherent in location sound)
- ❑ Equalised for special effect to reduce or add specific frequencies for design purposes
- ❑ Smoothed out for consistency within each scene and for transitions from one scene to the next
- ❑ Balance for transition across spool/reel changes
- ❑ **TIP:** If two elements play in the same frequency band at the same volume at the same time they fight each other for attention. The ear can perceive many different simultaneous sounds if they are separated by frequency.

Premixing

- ❑ Folding down of edited tracks to form premixes that can be played at unity across the mixing board prior to the Final Mix stage.
- ❑ Most of the Panning, Placement, Balancing and Equalisation is done at the premix stage
- ❑ Effects ie: delays, reverbs are added but kept separate in case of changes.
- ❑ Premix stems consist of Separate Dial, Fx, Atmosphere, and Foley premixes.
- ❑ As much separation of sound elements within premixes for changes in the Final Mix
- ❑ Music mixes delivered at Premix stage and a rough balance set
- ❑ Premix screening

Final Mix

- ❑ Premixes and Music are blended and weaved together to create the director's sonic vision
- ❑ Music is mixed and balanced to work with the other sound elements
- ❑ Additional Effects are added ie: delays reverbs etc
- ❑ Final Mix / Playback screening and Fixups
- ❑ Final Mix stems are now Dial, Music, and Effects
- ❑ A QC Check screening (Quality Control).

Printmastering

- ❑ A master of the final mix is produced for the optical/digital soundtrack for the DCP or audio file created for Layback to Digital Picture Master files.
- ❑ A Dolby consultant is sometimes present to oversee the technical alignment of the room and the mastering process.
- ❑ Some final tweaking of the mix can also be done at the Printmaster stage.
- ❑ QC Check screening/ Playback of Printmaster.

Delivery items

- ❑ Required by the production to fulfill contractual obligations
- ❑ Produced from the Final Mix Stems (Dial, Music, and Effects) to create cinema, international, domestic, airline, and television versions

NB: Mix intuitively, By ear, with regard to the Image. Sound enhances.

Dynamics, EQ, Effects, Placement & Panning

DYNAMICS & EQ

Dynamics and EQ should always be used with discretion, and always applied by ear. There is no stock standard, fix it solution for gating unwanted background noise or for compressing unevenly spoken dialogue. When working with film sound you must always be sure to enhance, not degrade the sounds you are working with.

When you first start to use these amazing tools, it is almost always a situation of going from one extreme to another, ie: either completely destroying the very sound you are trying to rectify or being too light-handed.

Either way, you have to start somewhere and it takes many hours to train your ears to listen for different frequencies, and to know which solutions to apply to the different problems you may encounter.

Most commonly used in film sound:

Compression

A compressor reduces the difference between the loudest and quietest parts of a piece of sound. It's used to even out and contain the volume or dynamic range of the audio signal, and can be used to "tighten up", "add punch" or increase "perceived" loudness to sounds.

GAIN: (gain makeup) HOW MUCH LEVEL ON OUTPUT

THRESHOLD: AT WHAT LEVEL DOES IT START WORKING

RATIO: HOW HARD IS IT COMPRESSING

ATTACK: HOW FAST DOES IT START WORKING

RELEASE: HOW QUICKLY OR SLOWLY DOES IT LET GO

KNEE: SMOOTHS OUT THE THRESHOLD POINT

Limiting

A limiter is a compressor with a fixed ceiling. Once the threshold is set, the audio signal is prevented from passing that point.

Gate

A gate is a device used to close down the audio path during periods of low signal level, when only noise is present, i.e. hiss, camera noise, etc.

A gate is just that – A GATE – it opens and closes'. It is used to get rid of unwanted noise and allow only the wanted sound to get through, set by the threshold.

THRESHOLD: AT WHAT LEVEL DOES IT START WORKING

ATTACK: HOW QUICKLY THE GATE OPENS

HOLD: HOW LONG DOES THE GATE STAY OPEN

DECAY: HOW QUICKLY THE GATE SHUTS

RANGE: HOW FAR IT OPENS OR CLOSES

EQ (Frequency)

Equalisation allows you to alter the timbre of the audio signal – Specific frequencies (wanted or unwanted audio) can be selected and Cut or Boosted.

Always equalise in context with the surrounding sounds.

- ❑ LOW FREQ = 20 – 200 kHz
- ❑ MID FREQ = 200 – 4000 kHz
- ❑ HIGH FREQ = 4000 – 20,000 kHz

- C = shelf - everything above this freq is boosted or cut
- O- = bell – or parametric equaliser
- o = shelf – everything below this freq is boosted or cut

EQ – 10 BANDS - (This is a guide. Use your own descriptions to define the range and what it sounds like to you personally).

- ❑ 60 Hz Subby, Whoofy, Rumbly - LOW
- ❑ 120 Hz Chesty, Boomy, Thundery,
- ❑ 250 Hz Muddy, Hollow – LOW MID
- ❑ 800 Hz Nasal, Honky, Boxy – MID RANGE
- ❑ 1500 Hz Megaphone, Transistor radio, bitey
- ❑ 3000 Hz Crack (snare drum), Vocal thinning/thickening, Tinny
- ❑ 4000 HZ Presence – LO High, Defines and clarifies voice
- ❑ 5000 Hz Kick (Heavy metal-clicky), Lo Hats, Vocal Sibilance,
- ❑ 8000 Hz VOICE – ESSE (Sibilance), Crisper, Grabby, Sharp
- ❑ 10,000 Hz Hi-Hats, – little Vocal, shimmer - HIGH
- ❑ 15,000 Hz Airy, glassy, thin, zing, bright, sheen metallic

De-esser

Commonly used to “clean up” sibilant vocal sounds. Sibilance is the air passing between the teeth when the letters “s” and “t” are spoken. Situated between 2500 and 10,000 Hz in the frequency range. It is prominent in closely recorded dialogue (used in film) and de-essing the dialogue helps to smooth out the high-pitched and bitey unwanted sound.

EQ Tips

from “*The Six Elements of a Film Mix*”

- ❑ To make an effect stick out, brighten it up.
- ❑ To make it blend in, filter out the highs.
- ❑ If the part is busy - roll off the low end of the effect to make it blend in.
- ❑ If the part is open, add a low end to the effect to fill in the space.
- ❑ If the source part is mono and panned hard to one side make one side of the stereo effect brighter and one side darker.

- ❑ Layer reverbs by the frequency with the longest being the brightest and the shortest being the darkest.
- ❑ Try not to pan reverbs hard left and hard right.
- ❑ Return the reverb in mono and pan accordingly - all reverbs needn't be returned in stereo.
- ❑ Get bigness from reverbs and depth from delays or vice versa.
- ❑ Use a bit of the longest reverb on all major elements of a mix to tie all the environments together.
- ❑ If delays are timed to the tempo of a track they add depth without being noticeable.
- ❑ If you want a delay to stick out make sure that it's not timed to the tempo of a track.
- ❑ Reverbs work better if they are timed to the tempo - if the reverb dies just before the next hit occurs.

EFFECTS

are used for spatial placement and time modification in synchronisation with the Picture.

Reverb

Used to simulate different acoustical environments. From Large Church halls with wooden floors to small slappy bathrooms with tiled walls.

It is used to place RAW ADR and or location dialogue, Fx into a designated space so that the voice / Fx appear to “breathe” in the environment.

It creates a spatial palette that is perceived by the listener to fit intrinsically with the image on the screen.

Delay/Echo

A time modifying effect that can be set to work in rhythm with the original audio signal, in very short or long increments.

Modulation Effects

Flanging, Phasing, and Chorus effects are used to fatten or broaden or creatively “weird out” the original sound source.

PLACEMENT and PANNING

Particular attention is paid to the placement and panning of sounds across the screen to “attach” the sound to characters or events on screen. Sound follows moving objects anywhere in the cinematic environment.

Correct panning makes the mix sound bigger, wider, and deeper. It also creates excitement by adding movement to the track and adding clarity to an idea.

TIP: Keep important elements in the centre of the mix. (IE: almost all Dialogue will be Centre, as it is considered KING in the hierarchy of film sound).

L = Left

C = Centre (also sounds can be centre left and centre right)

R = Right

LS = Left Surround

RS = Right Surround

Lfe = Sub Bass Channel

R/R= Right Rear

L/R= Left Rear

Back to front spatial placement can also be achieved.

Dimension

Creating acoustic environments

Dimension can be captured in recording but usually must be enhanced or created by adding effects such as reverb, delay, or modulated delays such as chorusing or flanging.

- ☐ The wetter the fx is, the further away it will sound
- ☐ Long delays and reverbs push the sound away if they are loud enough
- ☐ Smaller reverbs and short delays make things sound bigger
- ☐ Reverb decays under one second and delays under 100 milliseconds tend to create acoustic space around a sound
- ☐ Motion is created through a subtle combination of panning, equalization, and effects processing.